

Augmented Reality-Assisted Guide Plate Positioning in Lumbar Surgery Using Apple Vision Pro

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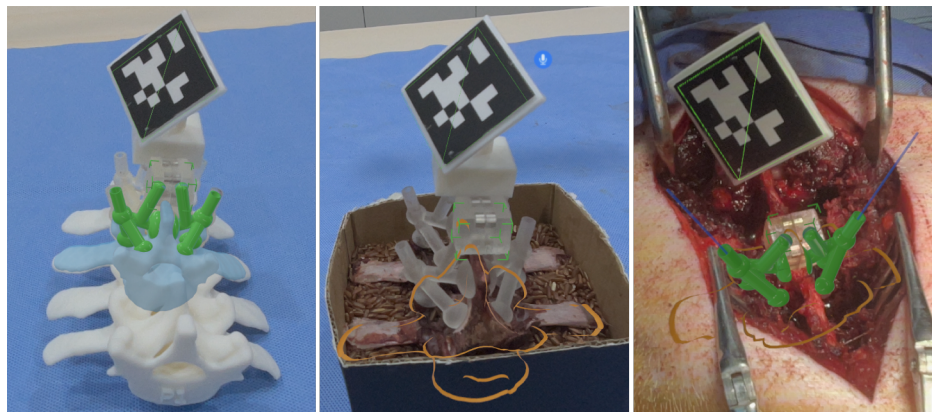


Figure 1: AR-assisted guide plate positioning in lumbar surgery using Apple Vision Pro. The method was evaluated on 3D-printed lumbar phantom (left), an *ex vivo* lumbar specimen from a pig (middle) and *in vivo* on two pigs during a surgery (right).

ABSTRACT

We propose an AR system using Apple Vision Pro (AVP) to assess the accuracy of guide plate positioning in lumbar surgery. It leverages preoperative CT data and integrates fiducial marker tracking to automate the initial alignment of a virtual vertebra with the intraoperative anatomy. Additionally, we developed a graphical interface for the AVP platform, which allows manual adjustment of the pose and visualization of the virtual vertebra. The system was tested on 3D-printed lumbar phantoms, an *ex vivo* pig lumbar specimen, and two *in vivo* pig surgeries. Results demonstrated accurate guide plate alignment and superior visualization, validating the potential of AR systems in improving surgical precision in lumbar procedures.

Index Terms: Augmented reality, lumbar surgery, surgical guide plate, 3D printing, Apple Vision Pro.

1 INTRODUCTION

Lumbar pedicle screw fixation is a critical surgical procedure for treating spinal disorders, where accurate placement of screws within the vertebrae is essential for spinal stabilization and fusion [1]. To achieve this objective, 3D-printed guide plates are increasingly used, crafted from preoperative CT scans to conform to the individual's vertebra geometry and provide patient-specific surgical guidance [2]. Despite these advancements, the presence of residual soft tissue can lead to deviations from the preoperative models, complicating the accurate positioning of the guide plate dur-

ing surgery. Consequently, physicians often rely on radioactive X-ray imaging to verify the guide plate's positioning, ensuring that it aligns with the surgical plan.

Augmented reality (AR) emerges as a promising solution for surgical navigation. Head-mounted devices (HMD), such as the Microsoft HoloLens, have been used in various studies that demonstrate the feasibility of AR in various procedures [3], especially in spine surgeries [4]. However, despite the prevalent use of HoloLens in these studies, the hardware's current capabilities still struggle to meet the stringent requirements for accuracy and reliability in tracking and registration essential for clinical practice [3].

Apple Vision Pro (AVP) is an AR headset launched by Apple Inc. in February 2024¹. It streams video from external cameras onto high-resolution internal screens, providing a realistic visual experience with seamless real-virtual alignment. AVP's key feature is its low-latency, precise tracking of eye and hand movements, facilitating intuitive interactions with virtual objects in 3D space. Powered by visionOS, it offers developers tools to create immersive AR apps. While AVP's potential in surgical AR applications seems promising, its efficacy in this context remains largely unexplored.

2 METHODOLOGY

In this study, we explore the feasibility of AR-assisted guide plate positioning in lumbar surgery using AVP, marking what we believe to be the first investigation of AVP's potential in orthopedic surgery. Figure 2 outlines the proposed method, which was tested on three pigs. Initially, CT scans were obtained, and then segmentations of the lumbar vertebrae were obtained for the design of guide plates to fit each vertebra's surface. A marker platform, featuring an attachment plane for a fiducial marker, was created to connect with the guide plates. Both the lumbar phantoms and the associated devices were 3D-printed. During the experiments, AVP tracked the fiducial

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¹<https://www.apple.com/apple-vision-pro/>

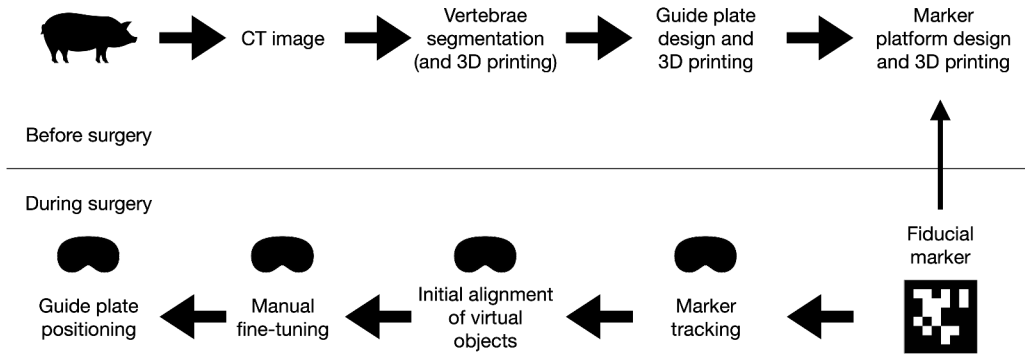


Figure 2: An overview of the AR-assisted guide plate positioning in lumbar surgery.

marker to overlay virtual bones and guide plates onto the real vertebrae or 3D-printed phantoms, aligning them based on their spatial relationships in the CT data. To accommodate intraoperative anatomical variations, a graphical user interface was designed for AVP to adjust the virtual objects' position and orientation, ensuring precise alignment during surgeries.

3 RESULTS

The experiments aimed to assess the accuracy of guide plate positioning. Initially, virtual vertebrae were aligned with real ones using both initial alignment and manual fine-tuning with a 4 cm fiducial marker. The virtual guide plate's pose was then determined from the spatial relationship between the vertebra and guide plate, as designed. During the alignment, the marker platform was mounted on a different vertebral segment from the segment where the guide plate of interest is. Once aligned, the virtual guide plate acted as a reference for real guide plate placement in surgery. To evaluate the alignment accuracy, we conducted three distinct experiments: using 3D-printed lumbar phantoms, *ex vivo* lumbar specimens, and *in vivo* lumbar vertebrae in a pig surgery (See Fig. 1).

In the 3D-printed phantom experiment, as the phantom's design was derived directly from the segmentation of the lumbar spine in the preoperative CT data, virtual objects were positioned using only initial alignment without manual fine-tuning. Alignment errors were assessed from the caudal and left lateral views by measuring the distance between real and virtual objects. As shown in Table 1, the average alignment error for the guide plate was under 3 mm, with tracker errors exceeding those of the guide plate, suggesting a reduction in error during transformation.

The *ex vivo* lumbar specimen experiment involved separate vertebral segments from L2 to L5, extracted from a pig after a surgery. We manually assembled these into four dual-lumbar segments (see Table 2), namely (L3, L2), (L2, L3), (L3, L4), and (L4, L5), thus their spatial relationships may be altered from preoperative CT data. After automatic initial alignment, we performed manual fine-tuning on the vertebrae. As shown in Table 2, manual fine-tuning significantly reduced alignment errors in all segments, compensating for discrepancies not addressed by the automatic initial alignment.

To assess the *in vivo* precision of AR-assisted guide plate positioning in clinical surgery, we performed pedicle screw placements on two pigs, utilizing both initial alignment and manual fine-tuning. Accuracy was evaluated by capturing X-ray fluoroscopic images with a mobile C-arm from anteroposterior and lateral views, which were compared to synthetic X-ray images reconstructed from pre-operative strew trajectory planning based on CT data using the same C-arm pose. We measured angular and entry point errors for 12 implanted screws. The average angular error was $1.29^\circ \pm 1.09^\circ$, and the average entry point error was 2.34 ± 1.37 mm.

Table 1: The average alignment error (mm) of the guide plate and the marker tracker from the caudal and the left views for lumbar segments from L2 to L5 on 3D-printed phantoms of two pigs.

Subject	Pig 1		Pig 2	
	Guide plate	Marker	Guide plate	Marker
View	Caudal / Left			
Error	1.52 / 2.13	2.79 / 3.40	1.07 / 1.49	3.10 / 2.67

Table 2: The average alignment error (mm) of the guide plate for the specimen lumbar segment from L2 to L5 before and after fine-tuning (FT) on a lumbar specimen. (A, B) means that the alignment of lumbar segment B is using the marker on segment A.

Segments	(L3, L2)	(L2, L3)	(L3, L4)	(L4, L5)
View	Caudal / Left			
Before FT	9.55 / 13.97	4.21 / 3.40	3.53 / 7.54	2.15 / 8.34
After FT	2.34 / 3.56	1.69 / 1.47	2.39 / 3.73	1.03 / 3.30

4 CONCLUSION

In conclusion, our system leveraging Apple Vision Pro facilitates AR-assisted guide plate positioning in lumbar surgery. It utilizes preoperative CT scans and 3D printing to create patient-specific guide plates and marker platforms. The system achieves initial alignment through fiducial marker tracking, followed by a fine-tuning process to accommodate deviations from preoperative planning. In the experiments, the system was evaluated across 3D-printed lumbar phantoms, *ex vivo* pig specimens, and *in vivo* surgeries on two pigs, and demonstrated high accuracy in guide plate alignment, particularly after fine-tuning. These findings highlight the system's potential for accurate guide plate positioning and surgical navigation in lumbar surgeries.

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